

1. Summary

Given a star player, identify which kinds of players complement their playing style, using unsupervised player-style clustering (k-means and Gaussian Mixture Models) on 2016-2024 NBA stats. The goal is to give teams a data-driven way to think about which player types to keep, trade for, or develop.

2. Target / Task

- **Task type:** Unsupervised learning (clustering), followed by a rule-based complementarity step.
- **Unit of analysis:** One player-season (a player's regular-season stat line for one year).
- **Step 1, play-style clusters:** Group player-seasons into play-style types using k-means and GMM.
- **Step 2, complementarity:** For a chosen star, define complementary clusters by an explicit written rule (for example, a high-usage scorer pairs with low-usage, high-efficiency spacers, rebounders, or playmakers). The rule is written down before looking at results.
- **Step 3, validation:** Test whether teams that pair a star type with complementary clusters have better team outcomes (net rating or win %) than teams that don't, on held-out seasons.
- **Baselines:** (a) random cluster assignment with the same k, (b) traditional position labels (PG/SG/SF/PF/C), (c) k-means with k=2 as a trivial reference. A one-group baseline is not used because silhouette is undefined for a single cluster.

3. Data Source and License

- **Dataset:** Source: NBA Stats (1980-2024) by rodneycarroll78 on Kaggle <https://www.kaggle.com/datasets/rodneycarroll78/nba-stats-1980-2024>
- **Upstream source:** Basketball Reference (Sports Reference LLC).
- **Files used:** Advanced.csv and Team Summaries.csv, both in data/raw/nba_stats/. We created a cleaned CSV called cleaned_nba.csv in data/processed/.
- **License:** CC0: Public Domain
- **Personal data:** Public professional statistics only. No contact details, health data, or private information.
- **Retrieved on:** Oct 2, 2026
- **Version/update noted on Kaggle:** 2 years ago

4. Data Card

Item	Detail
Name	NBA Stats (1980-2024)
Creator/curator	Rodney Carroll (Kaggle); stats from Basketball-Reference
Coverage used	2016 through 2024 seasons
Granularity	One row per player per season (a mid-season trade produces one row per team plus a combined TOT row)
Approx. size after filtering	3,222 player-season rows $\times$ 9 columns
Selected features	Usage % (usg_percent), true shooting % (ts_percent), 3-point attempt rate (x3p_ar), assist % (ast_percent), and rebound % (trb_percent).
Identifiers	Player, season, team, and minutes played
Known limitations	Box-score stats miss off-ball value, defensive impact, and lineup context. Positions are Basketball Reference's labels. Early-career and low-minute players have noisy stats. Era effects (pace, three-point volume) shift stat distributions across seasons.
Missing values	Missing values were present in several selected variables before cleaning. Rows with missing values in the required analysis columns were removed. In particular, usg_percent had 5,640 missing values, x3p_ar had 6,490, ast_percent had 2,725, trb_percent had 3,709, mp had 1,083, and ts_percent had 136. After removing rows with missing required values, the data were filtered to 2016–2024 and to player-seasons with at least 500 minutes

5. Sample Inputs and Outputs

Input (one row per player-season, after cleaning; the five selected features are before clustering):

Player	Season	Team	MP	usg_percent	ts_percent	x3p_ar	ast_percent	trb_percent
A.J. Green	2024	MIL	614.0	15.2	0.617	0.862	6.4	5.7
Aaron Gordon	2024	DEN	2297.0	17.8	0.607	0.193	15.2	11.4
Aaron Holiday	2024	HOU	1269.0	16.7	0.578	0.520	15.4	5.2

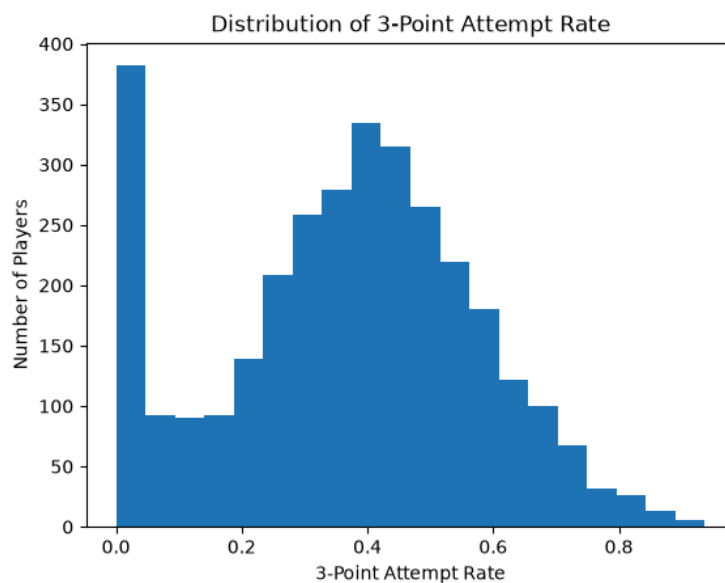
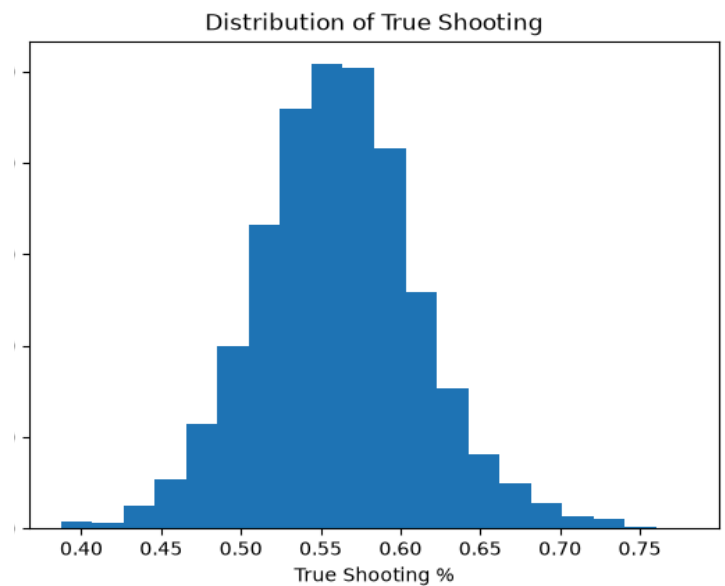
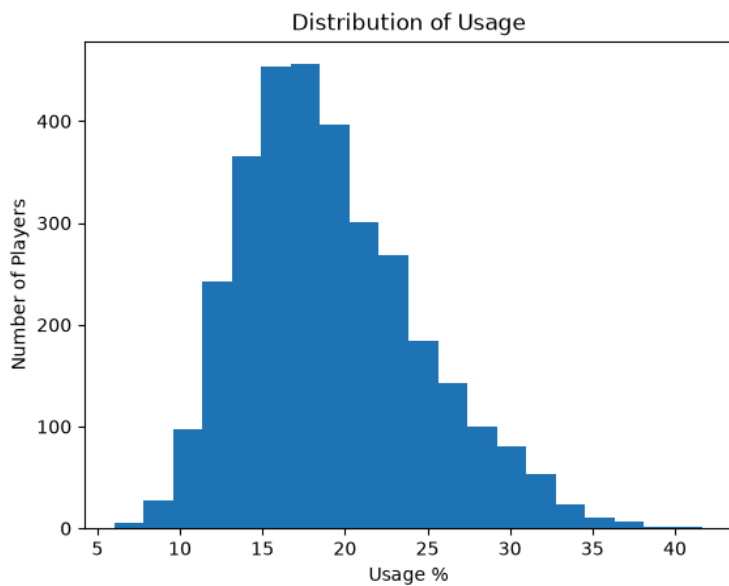
Output: cluster assignments produced by K-Means and GMM:

player	season	kmeans_cluster	gmm_cluster	gmm_probability
A.J. Green	2024	2	1	0.983291
Aaron Gordon	2024	1	2	0.647685
Aaron Holiday	2024	2	3	0.953446
Aaron Nesmith	2024	2	1	0.966124
Aaron Wiggins	2024	2	1	0.984818

6. Initial EDA Plan and Results

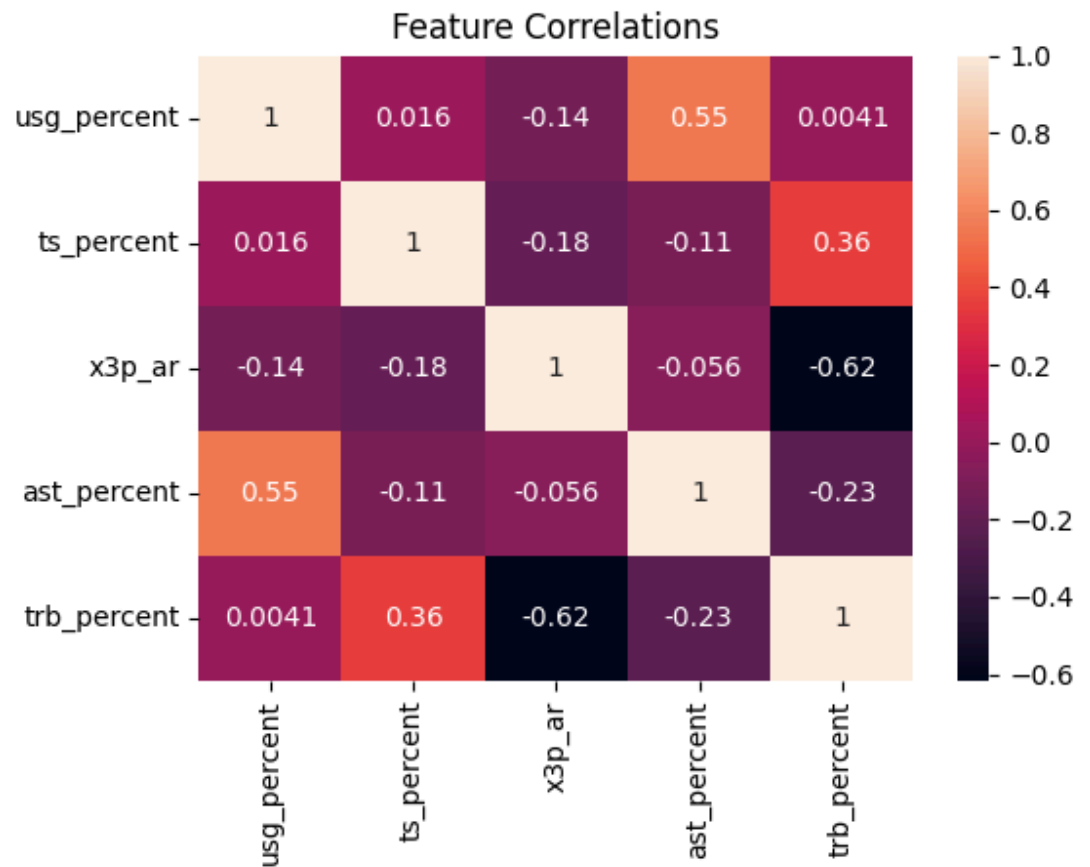
1. **Shape and types:** The original dataset had 31,870 rows and 32 columns and covered the 2016–2024 seasons. After cleaning, 4,789 player-seasons remained before the minutes filter, and 3,222 remained in the final dataset.
2. **Missing and invalid values:** Several features had missing values, so rows with missing values were removed. There were no negative minutes or other negative statistics. Six values had a true shooting percentage above 1, but they were not included in the final dataset.

3. **Minutes filter:** Players with fewer than 500 minutes in a season were removed. This removed 1,567 player-seasons, leaving 3,222.
4. **Duplicates and traded players:** For players who played for multiple teams in one season, the combined TOT row was kept, and the individual team rows were removed. No exact duplicate rows were found.
5. **Feature distributions:** Histograms were created for the selected features. The features had different ranges and distributions, so they were standardized before clustering.



6. Correlation matrix:

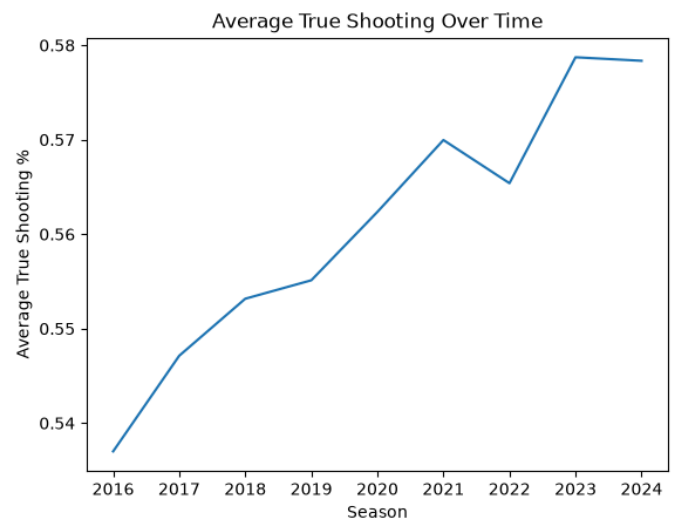
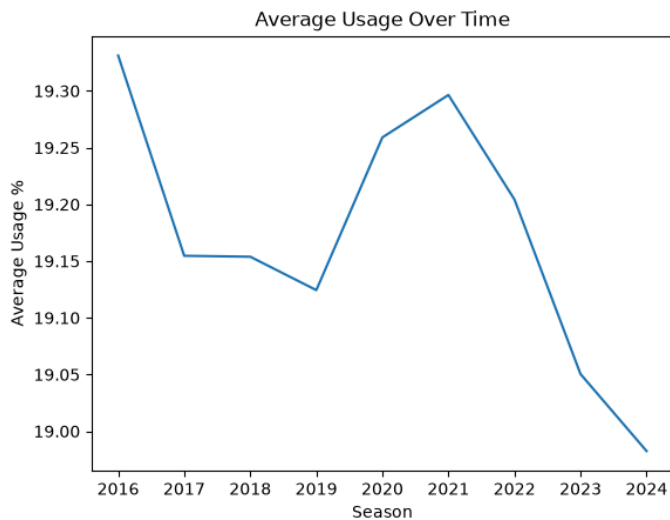
a. Figure 4. Correlation Between Selected Features



No pair of features had a correlation greater than 0.70 in absolute value.

Therefore, all five selected features were kept.

7. **Era drift:** Three-point attempt rate increased from about 0.282 in 2016 to 0.402 in 2024. True shooting percentage also increased. The features will be standardized before clustering.



7. Train / Validation / Test Strategy

Clustering has no labels, but we still need to choose k , the model, and the feature set without using our final data, and we need unseen seasons so we can test.

- **Split by season:** we split by season so the test seasons come after the training seasons, which avoids training on future data.
 - **Train:** 2016-17 to 2020-21. We fit the scaler and the clustering models here.
 - **Validation:** 2021-22 to 2022-23. We choose k and the feature set using silhouette, stability, and interpretability.
 - **Test:** 2023-24. We assign players to the frozen clusters and report final metrics once.
- We fit the scaler on train only and then apply it to validation and test.
- We set random seeds and use multiple k -means initializations. We report results as mean and spread across seeds.
- The team-outcome check in Section 8 uses the same split. If we change the split, we will document it.

8. Metrics

- **Primary metric:** Silhouette score on the held-out (test) seasons using clusters fitted on train, compared against the random-assignment and position-label baselines. The k and model are chosen on validation only.
- **Success criterion:** Silhouette clearly above the random baseline and at least as high as position labels. We will set how large the margin over random must be using validation seasons and before looking at test results. Because random labels score near zero, beating them is a weak bar, so beating position labels is the meaningful one. Also, because player statistics vary we expect modest values, and we will judge them relative to the baseline.
- **Secondary metrics:**
 - Calinski-Harabasz and Davies-Bouldin scores.
 - Adjusted Rand index between k-means and GMM, plus stability across seeds and seasons.
 - **Downstream validity:** correlation (or a simple regression) between a team's complementarity score and net rating or win %, controlling for total talent (for example, team BPM). This is the evidence that "complementary" means something.
- **Qualitative check:** cluster profiles and named archetypes reviewed by hand.

9. Risks and Mitigations

Risk	Mitigation
Dataset licensing or usage restrictions: The project uses a dataset whose statistics originate from Basketball-Reference, so there may be restrictions or conditions on redistributing the underlying data even if the Kaggle dataset is listed as CC0.	We will document the dataset source, license, retrieval date, and upstream source in the project documentation. The raw dataset will not be redistributed through the repository if its terms do not permit it. The repository will instead provide instructions for obtaining the data and reproducing the preprocessing steps.
Clusters may identify similar players rather than complementary players: Clustering naturally groups players who have similar statistical profiles. However, the main research question is about finding players whose skills complement a star, which is different from simply finding similar players.	We will define the complementarity rule before examining the final results. For example, a high-usage star may benefit from players with strong shooting efficiency, three-point shooting, rebounding, or playmaking but lower usage. The resulting recommendations will then be evaluated using team-level outcomes rather than assuming that different clusters automatically mean better fits.

Clustering may not produce meaningful player archetypes:

NBA player skills exist on a spectrum, meaning there may not be clearly separated groups of players. This could result in low silhouette scores or unstable clusters.

We will report clustering quality honestly using silhouette, Calinski-Harabasz, and Davies-Bouldin scores. We will also compare k-means and GMM results, examine cluster profiles, and compare them with the position-label baseline. If the clustering structure is weak, we will state that clearly rather than forcing an interpretation.

Choice of number of clusters may affect recommendations:

Different values of k can produce substantially different player groups and therefore different complementary recommendations.

The number of clusters will be selected using the validation seasons rather than the final test season. Multiple values of k will be tested, and the final choice will consider clustering quality, stability, and whether the resulting clusters are interpretable.

Correlation does not imply causation:

A team with a high complementarity score may also have better players, coaching, injuries, or other advantages that explain its success.

Team talent will be included as a control when evaluating the relationship between complementarity and team performance. Results will be described as associations rather than evidence that a particular player type directly causes improved team performance.

Team success depends on factors that are not included in the dataset:

Box-score statistics cannot fully capture coaching, defensive schemes, chemistry, injuries, lineup combinations, off-ball movement, or other contextual factors.

These factors will be identified as limitations of the project. The model will be presented as an exploratory decision-support tool rather than a system capable of determining the ideal roster by itself.

Era differences may affect player statistics:

NBA playing styles have changed considerably, particularly in areas such as three-point attempts, pace, and shooting efficiency. A player's statistical profile from one era may therefore not be directly comparable with a player from another era.

We will examine feature distributions across seasons and fit the scaler on training seasons. We list the remaining era drift as a limitation, and may repeat the analysis with season standardization as a sensitivity check.

Small-minute players can have unreliable statistics:

Players with very limited playing time may have unusually high or low statistics that do not accurately represent their actual playing style.

Traded players may appear multiple times in a season:

A player who plays for multiple teams can have separate team rows as well as a combined TOT row. Counting these rows separately could give the same player-season disproportionate influence.

Selected features may not capture all aspects of a player's style:

Usage, shooting, assists, and rebounding describe important parts of a player's statistical profile but do not fully represent defense, off-ball movement, screening, or other valuable skills.

Feature selection could unintentionally bias the clusters: Choosing features because they produce better-looking clusters after seeing the results could lead to overfitting or overly optimistic conclusions.

Data leakage between seasons:

A player may appear in multiple seasons, so randomly splitting player rows could allow information about the same player to appear in both training and testing data.

K-means and GMM may produce inconsistent results:

The two algorithms make different assumptions about cluster structure, so

We will remove player-seasons with fewer than 500 minutes. This reduces the effect of extremely small samples while retaining players who had meaningful NBA playing time.

For players who were traded, we will keep the combined TOT season and remove the individual team rows. This ensures that each player-season contributes consistently to the clustering process.

We will document the selected features and explain what each contributes to the player-style representation. Additional play-by-play or shooting data could be explored as a future extension if the current features do not adequately distinguish player styles.

Feature selection and the complementarity rule will be established before evaluating the final test results. Feature-set experiments will be conducted during validation, and the final test set will remain untouched until the final evaluation.

The project will use season-based train, validation, and test splits. The scaler will also be fitted only on the training data and then applied unchanged to validation and test data.

We will compare the cluster assignments using measures such as Adjusted Rand Index and examine stability across different random seeds and seasons. Differences between models will be

they may assign the same player to different groups.

GMM probabilities may indicate uncertainty:

Unlike k-means, GMM can assign a player a probability of belonging to different clusters. Players near the boundary between clusters may therefore be difficult to classify confidently.

The complementarity rule could be subjective:

Deciding that one cluster complements a star more than another could introduce researcher bias.

A statistically complementary player may not be a realistic roster fit:

Salary, contracts, availability, team chemistry, injuries, and positional requirements are not fully represented by the clustering model.

Player statistics may not represent team fit:

A player's individual statistics can look complementary while their actual combination with a star does not improve team performance.

Weak relationship between complementarity and team success:

Even if clusters are statistically meaningful, the complementarity score may show little relationship with team performance.

reported rather than treating one algorithm as automatically correct.

GMM probabilities will be recorded and considered when interpreting player assignments. Recommendations involving players with highly uncertain cluster membership will be treated more cautiously.

The complementarity rule will be explicitly written before examining the results. The rule will be based on measurable statistical characteristics rather than manually selecting individual players after seeing the results.

The model will be described as identifying **statistical fits**, not predicting actual trades or acquisitions. Any player recommendations will be presented as potential candidates for further evaluation rather than definitive roster decisions.

The project will include a downstream validation step comparing team complementarity scores with team outcomes such as net rating or win percentage. This provides an additional test of whether the identified complementary types have practical value.

This will be treated as a valid research result rather than a failure. We will report cluster quality separately from downstream team-outcome performance and discuss possible explanations for an inconclusive relationship.

Over-interpretation of model recommendations:

Teams or readers could interpret the results as saying that a particular player is objectively good or bad for a star player.

The project will emphasize that the model identifies statistical patterns and associations. Results will be presented as exploratory decision support and not as a definitive verdict on any individual player.

Reproducibility issues:

Different random seeds, package versions, preprocessing choices, or data versions could produce different clustering results.

The repository will include a requirements.txt, fixed random seeds, documented preprocessing procedures, experiment settings, and instructions for obtaining the dataset. This will allow the experiments to be reproduced as closely as possible.

Computational limitations are unlikely but possible:

Although the dataset is relatively small, testing multiple cluster counts, feature sets, random seeds, and models could increase experimentation time.

Experiments will be performed using standard CPU-based tools such as pandas and scikit-learn. Since the final dataset contains only a few thousand player-seasons, no GPU or cloud computing should be required.

10. Feasibility and Compute Plan

- **Data size:** Roughly a few thousand player-seasons after filtering. Small tabular data.
- **Models:** scikit-learn KMeans and GaussianMixture, silhouette and related metrics, pandas, matplotlib, seaborn.
- **Compute:** A standard laptop CPU is sufficient. Fitting and scoring across a range of k should take seconds to a few minutes. No GPU or cloud needed.
- **Timeline:**
 1. Data download, cleaning, EDA. **(10/03/2026)**
 2. Feature selection, baselines, k-means and GMM, validation tuning. **(10/10/2026)**
 3. Complementarity rule and team-outcome validation. **(October 31 – November 20, 2026)**
 4. Test-set evaluation, write-up, repository cleanup. **(November 21 – December 4, 2026)**
- **Fallbacks:** If GMM is unstable, use fewer features or diagonal covariance. If the team-outcome check is inconclusive, report cluster quality and qualitative recommendations and say so plainly.
- **Feasibility verdict:** High. Small data, standard methods, low compute.

11. Github Link

https://github.com/AllenBlesson/NBA_ML